

Francisco Castro, PhD

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Education

- 2020 **PhD in Computer Science**, *Worcester Polytechnic Institute (WPI)*, Massachusetts, USA.
Dissertation: [Development of a Data-Grounded Theory of Program Design in HTDP](#)
Committee
◦ Kathi Fisler, PhD, *Advisor*, Brown University and Worcester Polytechnic Institute
◦ Daniel Dougherty, PhD, Worcester Polytechnic Institute
◦ Jacob Whitehill, PhD, Worcester Polytechnic Institute
◦ Brian Dorn, PhD, University of Nebraska Omaha
- 2014 **Master in Information Technology**, *Ateneo de Manila University*.
- 2010 **Bachelor of Science in Computer Science**, *University of the Philippines Los Baños*.

Research and Academic Appointments

- 2025 – present **Research Scientist – AI, Computing Ethics, and Human-Centered Technologies**, *Administration, Leadership, and Technology, Steinhardt School of Culture, Education, and Human Development, New York University, USA*.
- 2022 – 2024 **Computing Innovation Research Fellow – AI, AI Ethics, and Creative Computing**, *Administration, Leadership, and Technology, Steinhardt School of Culture, Education, and Human Development, New York University, USA*, cifellows2021.org.
- 2023 – 2025 **Visiting Researcher – Computing and AI Ethics Education and Wearable Technologies for Games and Learning**, *College of Information and Computer Sciences, University of Massachusetts Amherst, USA*.
- 2020 – 2023 **Postdoctoral Research Associate – Wearable Technologies for Games and Learning**, *College of Information and Computer Sciences, University of Massachusetts Amherst, Advanced Learning Technologies Laboratory, USA*, wearablelearning.org.
- 2017 – 2020 **Visiting Research Scientist**, *Department of Computer Science, Brown University, USA*.
- 2020 **Doctoral Research Assistant**, *Department of Computer Science, Worcester Polytechnic Institute, Advanced Learning Technologies Laboratory, USA*.
- 2015 – 2020 **Doctoral Research Assistant**, *Department of Computer Science, Worcester Polytechnic Institute, Computing Education Research Group, USA*.
- 2014 **Teaching Assistant**, *Department of Computer Science, Worcester Polytechnic Institute, USA*, [Teaching Assistant of the Year Awardee](#).
Courses taught:
◦ CS1101: Introduction to Program Design (*Fall 2014*)
◦ CS2102: Object Oriented Design Concepts (*Fall 2014*)
- 2013 **Visiting Research Scientist**, [Augmented Human Communication Laboratory](#), *Nara Institute of Science and Technology (NAIST), Japan*.

2013 – 2014 **Instructor**, *Department of Information Systems and Computer Science, Ateneo de Manila University, Philippines.*

Courses taught:

- CS30: Introduction to Information Technology (*1st Semester 2013, 2014*)
- CS112: Structure of Programming Languages (*2nd Semester 2013*)
- CS187.1: Applications in Medicine and Public Health Informatics (*2nd Semester 2013*)
- CS280: Programming Languages and Paradigms (*Co-teach, Summer 2014*)

2011 – 2013 **Instructor**, *Institute of Computer Science, University of the Philippines Los Baños, Philippines.*

Courses taught:

- CMSC2: Introduction to Web Development
- IT1: Information Technology Literacy
- CMSC11: Introduction to Computer Science
- CMSC22: Object Oriented Programming
- CMSC123: Data Structures

Industry Appointments

2020 – 2021 **Senior Data and AI Advanced Application Engineering Analyst**, *Accenture, Boston, Massachusetts, USA.*

2018 **Human-Computer Interaction Research and UX Engineer Intern**, *Amadeus, Waltham, Massachusetts, USA.*

2016 **Data Science Content Development and Engineering Intern**, *DataCamp, Cambridge, Massachusetts, USA.*

2014 **User Experience Designer**, *Traffic Computing Research Group, Advanced Traffic and Pollution Monitoring and Analysis System, Ateneo de Manila University, Philippines.*

2013 – 2014 **Technical and Research Consultant**, *Social Computing Science Lab and Institute of Philippine Culture, Ateneo de Manila University, Philippines.*

2009 **Web Developer**, *La Salle Provincialate Communications and Resources, Philippines.*

Publications

Refereed Journal Articles

2023 Gayithri Jayathirtha and **Francisco Enrique Vicente Castro**. Disaggregating Asian Identities Through Case Studies of High School Students in Electronic Textiles Classrooms. *Sustainability*, volume 15, 2023. DOI: [10.3390/su152015128](https://doi.org/10.3390/su152015128) (PDF).

2022 Juho Leinonen, **Francisco Enrique Vicente Castro**, and Arto Hellas. Time-on-task metrics for predicting performance. *ACM Inroads*, volume 13, pages 42–49, May 2022. DOI: [10.1145/3534564](https://doi.org/10.1145/3534564) (PDF).

2022 Ivon Arroyo, Avery Harrison Closser, **Francisco Castro**, Hannah Smith, Erin Ottmar, and Matthew Micciolo. The WearableLearning Platform: A Computational Thinking Tool Supporting Game Design and Active Play. *Technology, Knowledge and Learning*, April 2022. DOI: [10.1007/s10758-022-09601-1](https://doi.org/10.1007/s10758-022-09601-1) (PDF).

2013 **Francisco Enrique Vicente G. Castro** and Jaderick P. Pabico. A Study on the Effect of Exit Widths and Crowd Sizes in the Formation of Arch in Clogged Crowds. *Philippine Computing Journal*, volume 8, pages 21–29, August 2013. DOI: [10.48550/arXiv.1506.08133](https://doi.org/10.48550/arXiv.1506.08133) (PDF).

- 2013 **Francisco Enrique Vicente G. Castro** and Jaderick P. Pabico. Microsimulations of Arching, Clogging, and Bursty Exit Phenomena in Crowd Dynamics. *Philippine Information Technology Journal*, volume 6, pages 11–16, 2013. DOI: [10.48550/arXiv.1506.07781](https://doi.org/10.48550/arXiv.1506.07781) (PDF) **Best Paper Award**.
- 2013 Ma Mercedes T. Rodrigo, Thor Collin S. Andallaza, **Francisco Enrique Vicente G. Castro**, Marc Lester V. Armenta, Thomas T. Dy, and Matthew C. Jadud. An Analysis of Java Programming Behaviors, Affect, Perceptions, and Syntax Errors among Low-Achieving, Average, and High-Achieving Novice Programmers. *Journal of Educational Computing Research*, volume 49, pages 293–325, October 2013. DOI: [10.2190/EC.49.3.b](https://doi.org/10.2190/EC.49.3.b) (PDF).

Refereed Conference Publications

- 2026 Emily Martinez, Yashi Patel, Adyan Chowdhury, Noel Chacko, **Francisco Castro**, and Stephen MacNeil. Navigating computing careers: Tiktok's potential role as an informal resource. In *Proceedings of the 57th ACM Technical Symposium on Computer Science Education V.1*, SIGCSE TS 2026, page 708–714, New York, NY, USA, 2026. Association for Computing Machinery. DOI: [10.1145/3770762.3772651](https://doi.org/10.1145/3770762.3772651) (PDF).
- 2025 **Francisco Enrique Vicente Castro**. Case studies on responsible computing: Speculative thinking and considerations in technology ecosystems. In *Proceedings of the 26th ACM Annual Conference on Cybersecurity & Information Technology Education*, SIGCITE '25, page 207–212, New York, NY, USA, 2025. Association for Computing Machinery. DOI: [10.1145/3769694.3771138](https://doi.org/10.1145/3769694.3771138) (PDF).
- 2025 Cass Scheirer, **Francisco Castro**, and Kayla DesPortes. Towards A Bigger, Brighter Room: Exploring the Affordances of a Tangible Computing Tool Within a Creative Computing Program for Youth. In *Proceedings of the 24th Interaction Design and Children*, page 1080–1085, New York, NY, USA, 2025. Association for Computing Machinery. DOI: [10.1145/3713043.3731534](https://doi.org/10.1145/3713043.3731534) (PDF).
- 2025 Allison Poh, Yuqian Shi, **Francisco Enrique Vicente Castro**, and Ivon Arroyo. Enhancing Teacher Support in Learning Technologies: A Human-Centered Approach with WearableLearning. In Alexandra I. Cristea, Erin Walker, Yu Lu, Olga C. Santos, and Seiji Isotani, editors, *Artificial Intelligence in Education. Posters and Late Breaking Results, Workshops and Tutorials, Industry and Innovation Tracks, Practitioners, Doctoral Consortium, Blue Sky, and WideAIED*, pages 53–60, Cham, 2025. Springer Nature Switzerland. DOI: [10.1007/978-3-031-99264-3_7](https://doi.org/10.1007/978-3-031-99264-3_7) (PDF).
- 2025 Ava Heinonen, Reetta Puska, **Francisco Castro**, Juha Sorva, Juho Leinonen, and Arto Hellas. Investigating students' programming plan knowledge with time-constrained code recall tasks. In *2025 IEEE Frontiers in Education Conference (FIE)*, pages 1–9, 2025. DOI: [10.1109/FIE63693.2025.11328482](https://doi.org/10.1109/FIE63693.2025.11328482) (PDF).
- 2025 Mohammad Hadi Nezhad, **Francisco Enrique Vicente Castro**, Eugene Mak, Peter J. Haas, Danielle Allessio, Leon Osterweil, Injila Rasul, Heather Conboy, and Ivon Arroyo. Embedding Ethical Awareness in Computer Science and AI Education: The PEaRCE Approach to Responsible Computing. In Alexandra I. Cristea, Erin Walker, Yu Lu, Olga C. Santos, and Seiji Isotani, editors, *Artificial Intelligence in Education*, pages 135–149, Cham, 2025. Springer Nature Switzerland. DOI: [10.1007/978-3-031-98414-3_10](https://doi.org/10.1007/978-3-031-98414-3_10) (PDF).
- 2025 Nick Bryan-Kinns, Shuoyang Zheng, **Francisco Castro**, Makayla Lewis, Jia-Rey Chang, Gabriel Vigliani, Terence Broad, Michael Paul Clemens, and Elizabeth Wilson. Xaixarts manifesto: Explainable ai for the arts. In *Proceedings of the Extended Abstracts of the CHI Conference on Human Factors in Computing Systems*, CHI EA '25, New York, NY, USA, 2025. Association for Computing Machinery. DOI: [10.1145/3706599.3716227](https://doi.org/10.1145/3706599.3716227) (PDF).
- 2025 Jill Beale, **Francisco Castro**, Xiaomeng Huang, Cass Scheirer, Chuqing Lyu, Yuning Gao, and Kayla DesPortes. Critical thinking habits of mind in media arts learning. In A. Rajala, A. Cortez, R. Hofmann, A. Jornet, H. Lotz-Sisitka, and L. Markauskaite, editors, *Proceedings of the 19th International Conference of the Learning Sciences - ICLS 2025*, pages 610–618, 2025. DOI: [10.22318/icls2025.816802](https://doi.org/10.22318/icls2025.816802) (PDF).

- 2024 **Francisco Enrique Vicente Castro**, Earl Huff, Amber Solomon, and Briana Bettin. Bringing Our Full Selves Into Computing: Designing, Building, and Fostering Equitable Computing Education Communities. In *Proceedings of the 2024 on Innovation and Technology in Computer Science Education V. 1*, ITiCSE 2024, page 136–142, New York, NY, USA, July 2024. Association for Computing Machinery. DOI: [10.1145/3649217.3653550](https://doi.org/10.1145/3649217.3653550) (PDF).
- 2024 **Francisco Enrique Vicente Castro**, shuang cai, Vera Liqian Zhong, and Kayla DesPortes. Body and Code: A Distributed Cognition Exploration Into Dance and Computing Learning. In *Proceedings of the 16th Conference on Creativity & Cognition*, C&C '24, page 196–210, New York, NY, USA, June 2024. Association for Computing Machinery. DOI: [10.1145/3635636.3656206](https://doi.org/10.1145/3635636.3656206) (PDF).
- 2024 **Francisco Castro** and Kayla DesPortes. Exploring, Co-Creating, and Reflecting on Embodied Creative Dance Computing Learning Activities for AI Education. In *Proceedings of the American Educational Research Association 2024 Annual Meeting*, AERA '24. American Educational Research Association, April 2024. (PDF: OSF - <https://osf.io/gwe27>).
- 2024 **Francisco Castro** and Kayla DesPortes. Dance and Creative Computing as Spaces for Critically Engaging and Learning About Artificial Intelligence/Machine Learning. In *Proceedings of the American Educational Research Association 2024 Annual Meeting*, AERA '24. American Educational Research Association, April 2024. (PDF: OSF - <https://osf.io/adt57>).
- 2024 Cass Scheirer, Vera Zhong, Sakshi Kulkarni, Mashiyat Zaman, Kayla Desportes, and **Francisco Enrique Vicente Castro**. touchBase: A Tangible Programming Language for Physical Computing. In *Proceedings of the 16th Conference on Creativity & Cognition*, C&C '24, page 464–469, New York, NY, USA, June 2024. Association for Computing Machinery. DOI: [10.1145/3635636.3664253](https://doi.org/10.1145/3635636.3664253) (PDF).
- 2024 Adaurennaya C. Onyewuenyi, **Francisco Castro**, Nichole Garcia, Susan Brown Trinidad, Yoon Jeon Kim, Danielle Espino, and Brendan Eagan. Methodological Code Switching: Using Criticality to Expand Quantitative Ethnography's Bounds. In *Proceedings of the 2024 International Conference on Quantitative Ethnography*, ICQE '24, 2024. (PDF: International Society for Quantitative Ethnography - https://www.qesoc.org/images/pdf/ICQE24_Supplement_Proceedings.pdf#page=64).
- 2024 Luis Morales-Navarro, Yasmin Kafai, Ha Nguyen, Kayla Desportes, Ralph Vacca, Camillia Matuk, Megan Silander, Anna Amato, Peter Woods, **Francisco Castro**, Mia Shaw, Selin Akgun, Christine Greenhow, and Antero Garcia. Learning about Data, Algorithms, and Algorithmic Justice on TikTok in Personally Meaningful Ways. In R. Lindgren, T. I. Asino, E. A. Kyza, C. K. Looi, D. T. Keifert, and E. Suárez, editors, *Proceedings of the 18th International Conference of the Learning Sciences - ICLS 2024*. International Society of the Learning Sciences, June 2024. DOI: [10.48550/arXiv.2405.15437](https://doi.org/10.48550/arXiv.2405.15437) (PDF).
- 2024 Sai Gattupalli, **Francisco Enrique Vicente G. Castro**, and Ivon Arroyo. Investigating Cultural Signatures in Game Designs of Middle School Students from Argentina, India, and the United States. In R. Lindgren, T. I. Asino, E. A. Kyza, C. K. Looi, D. T. Keifert, and E. Suárez, editors, *Proceedings of the 18th International Conference of the Learning Sciences - ICLS 2024*, pages 1518–1521. International Society of the Learning Sciences, June 2024. DOI: [10.22318/icls2024.138871](https://doi.org/10.22318/icls2024.138871) (PDF).
- 2024 Kayla DesPortes, Kathleen McDermott, Yoav Bergner, **Francisco Enrique Vicente Castro**, Sauda Musharrat, and Aakruti Lunia. DanceBits 'It tells you to see us': Supporting Dance Practices with an Educational Computing Kit. In *Proceedings of the Eighteenth International Conference on Tangible, Embedded, and Embodied Interaction*, TEI '24, New York, NY, USA, 2024. Association for Computing Machinery. DOI: [10.1145/3623509.3633350](https://doi.org/10.1145/3623509.3633350) (PDF).

- 2023 **Francisco Castro**, Sahitya Raipura, Heather Conboy, Peter Haas, Leon Osterweil, and Ivon Arroyo. Piloting an interactive ethics and responsible computing learning environment in undergraduate cs courses. In *Proceedings of the 54th ACM Technical Symposium on Computer Science Education V. 1*, SIGCSE 2023, page 659–665, New York, NY, USA, 2023. Association for Computing Machinery. DOI: [10.1145/3545945.3569753](https://doi.org/10.1145/3545945.3569753) (PDF).
- 2023 Injila Rasul, **Francisco Castro**, and Ivon Arroyo. Towards embodied wearable intelligent tutoring systems. In Claude Frasson, Phivos Mylonas, and Christos Troussas, editors, *Augmented Intelligence and Intelligent Tutoring Systems: 19th International Conference, ITS 2023, Corfu, Greece, June 2–5, 2023, Proceedings*, page 298–306, Berlin, Heidelberg, 2023. Springer Nature Switzerland. DOI: [10.1007/978-3-031-32883-1_26](https://doi.org/10.1007/978-3-031-32883-1_26) (PDF).
- 2023 Injila Rasul, Danielle Crabtree, **Francisco Castro**, Allison Poh, Sai Satish Gattupalli, Krishna Chaitanya Rao Kathala, and Ivon Arroyo. WearableLearning: Developing Computational Thinking Through Modeling, Simulation, and Computational Problem Solving. In *Proceedings of the 2023 International Conference of the Learning Sciences - ICLS 2023*. International Society of the Learning Sciences, May 2023. (PDF: OSF - <https://osf.io/d73yh>).
- 2023 Allison Poh, **Francisco Enrique Vicente Castro**, and Ivon Arroyo. Design Principles for Teacher Dashboards to Support In-Class Learning. In *Proceedings of the 2023 International Conference of the Learning Sciences - ICLS 2023*. International Society of the Learning Sciences, May 2023. (PDF: OSF - <https://osf.io/mxkjh>).
- 2023 Luis Morales-Navarro, Yasmin B. Kafai, **Francisco Castro**, William Payne, Kayla DesPortes, Daniella DiPaola, Randi Williams, Safinah Ali, Cynthia Breazeal, Clifford Lee, Elisabeth Soep, Duri Long, Brian Magerko, Jaemarie Solyst, Amy Ogan, Cansu Tatar, Shiyan Jiang, Jie Chao, Carolyn P. Rosé, and Sepehr Vakil. Making Sense of Machine Learning: Integrating Youth's Conceptual, Creative, and Critical Understandings of AI. In *Proceedings of the 2023 International Conference of the Learning Sciences - ICLS 2023*. International Society of the Learning Sciences, May 2023. DOI: [10.48550/arXiv.2305.02840](https://doi.org/10.48550/arXiv.2305.02840) (PDF).
- 2023 Michael J Johnson, **Francisco Enrique Vicente Castro**, Betsy DiSalvo, and Kayla DesPortes. Chronicles of exploration: Examining the materiality of computational artifacts. In *Proceedings of the 2023 ACM Conference on International Computing Education Research - Volume 1*, ICER '23, page 29–47, New York, NY, USA, 2023. Association for Computing Machinery. DOI: [10.1145/3568813.3600132](https://doi.org/10.1145/3568813.3600132) (PDF).
- 2023 Kayla DesPortes, **Francisco Enrique Vicente Castro**, and William Payne. AI Ethics Education in Dance and Creative Computing Spaces. In *Proceedings of the American Educational Research Association 2023 Annual Meeting*, AERA '23. American Educational Research Association, April 2023. (PDF: OSF - <https://osf.io/qz683>).
- 2022 **Francisco Enrique Vicente Castro**, Juho Leinonen, and Arto Hellas. Experiences with and lessons learned on deadlines and submission behavior. In *Proceedings of the 22nd Koli Calling International Conference on Computing Education Research*, Koli Calling '22, New York, NY, USA, 2022. Association for Computing Machinery. DOI: [10.1145/3564721.3564728](https://doi.org/10.1145/3564721.3564728) (PDF).
- 2022 Juho Leinonen, **Francisco Enrique Vicente Castro**, and Arto Hellas. Time-on-Task Metrics for Predicting Performance. In *Proceedings of the 53rd ACM Technical Symposium on Computer Science Education V. 1*, SIGCSE 2022, pages 871–877, New York, NY, USA, February 2022. Association for Computing Machinery. DOI: [10.1145/3478431.3499359](https://doi.org/10.1145/3478431.3499359) (PDF) **Best Paper Award**.

- 2022 Shiyan Jiang, Kayla DesPortes, Yoav Bergner, Helen Zhang, Irene Lee, Katherine Moore, Yihong Cheng, Beatriz Paris, Benjamin Walsh, Aaron Guggenheim, Bridget Dalton, Stacey Forsyth, Tom Yeh, Bitu Akram, Spencer Yoder, William Finzer, Jie Chao, Carolyn Rosé, William Payne, **Francisco Castro**, and Kathleen McDermott. Agents, Models, and Ethics: Importance of Interdisciplinary Explorations in AI Education. In *Proceedings of the International Conference of the Learning Sciences*. International Society of the Learning Sciences, June 2022. (PDF: [OSF - https://osf.io/b3s5e](https://osf.io/b3s5e)).
- 2022 Sai Gattupalli, **Francisco Enrique Vicente Castro**, Ivon Arroyo, Olivia Bogs, and Grace Seiche. Exploring Cross-cultural Differences in Educational Math Game Creation Using the WearableLearning Platform. In *Proceedings of the International Conference of the Learning Sciences*. International Society of the Learning Sciences, June 2022. (PDF: [OSF - https://osf.io/yq49e/files/kuhqd](https://osf.io/yq49e/files/kuhqd)).
- 2021 Juho Leinonen, **Francisco Enrique Vicente Castro**, and Arto Hellas. Does the Early Bird Catch the Worm? Earliness of Students' Work and its Relationship with Course Outcomes. In *Proceedings of the 26th Annual Conference on Innovation and Technology in Computer Science Education, ITiCSE '21*, Paderborn, Germany, June 2021. Association for Computing Machinery. DOI: [10.1145/3430665.3456383](https://doi.org/10.1145/3430665.3456383) (PDF).
- 2021 Ivon Arroyo, **Francisco Enrique Vicente Castro**, Hannah Smith, Avery Closser, and Erin Ottmar. Augmenting Embodied Mathematics Classrooms With Mobile Tutors. In *Proceedings of the American Educational Research Association 2021 Annual Meeting, AERA '21*. American Educational Research Association, April 2021.
- 2020 **Francisco Enrique Vicente Castro** and Kathi Fisler. Qualitative Analyses of Movements Between Task-level and Code-level Thinking of Novice Programmers. In *Proceedings of the 51st ACM Technical Symposium on Computer Science Education, SIGCSE '20*, pages 487–493, Portland, OR, USA, February 2020. Association for Computing Machinery. DOI: [10.1145/3328778.3366847](https://doi.org/10.1145/3328778.3366847) (PDF).
- 2018 Jun Rangie C. Obispo, **Francisco Enrique Vicente G. Castro**, and Ma. Mercedes Rodrigo. Incidence of Einstellung Effect among Programming Students and its Relationship with Achievement. In *Proceedings of the 1st Information and Computing Education Conference, ICE 2018*, October 2018. (PDF: [OSF - https://osf.io/g58qc/files/vy9mt](https://osf.io/g58qc/files/vy9mt)).
- 2017 **Francisco Enrique Vicente Castro**, Shriram Krishnamurthi, and Kathi Fisler. The Impact of a Single Lecture on Program Plans in First-year CS. In *Proceedings of the 17th Koli Calling Conference on Computing Education Research*, Koli Calling '17, pages 118–122, New York, NY, USA, 2017. ACM. DOI: [10.1145/3141880.3141897](https://doi.org/10.1145/3141880.3141897) (PDF).
- 2017 **Francisco Enrique Vicente Castro** and Kathi Fisler. Designing a Multi-faceted SOLO Taxonomy to Track Program Design Skills Through an Entire Course. In *Proceedings of the 17th Koli Calling Conference on Computing Education Research*, Koli Calling '17, pages 10–19, New York, NY, USA, 2017. ACM. DOI: [10.1145/3141880.3141891](https://doi.org/10.1145/3141880.3141891) (PDF) **Best Paper Award**.
- 2017 Kathi Fisler and **Francisco Enrique Vicente Castro**. Sometimes, Rainfall Accumulates: Talk-Alouds with Novice Functional Programmers. In *Proceedings of the 2017 ACM Conference on International Computing Education Research, ICER '17*, pages 12–20, New York, NY, USA, 2017. ACM. DOI: [10.1145/3105726.3106183](https://doi.org/10.1145/3105726.3106183) (PDF).
- 2016 **Francisco Enrique Vicente Castro** and Kathi Fisler. On the Interplay Between Bottom-Up and Datatype-Driven Program Design. In *Proceedings of the 47th ACM Technical Symposium on Computer Science Education, SIGCSE '16*, pages 205–210, New York, NY, USA, 2016. ACM. DOI: [10.1145/2839509.2844574](https://doi.org/10.1145/2839509.2844574) (PDF).
- 2015 **Francisco Enrique Vicente Castro**, Seth Adjei, Tyler Colombo, and Neil Heffernan. Building Models to Predict Hint-or-Attempt Actions of Students. In *Proceedings of the 8th International Conference on Educational Data Mining*. International Educational Data Mining Society, June 2015. (EDM 2015 – https://www.educationaldatamining.org/EDM2015/proceedings/edm2015_proceedings.pdf).

- 2014 Ma. Regina Estuar, Dennis Batangan, Andrei Coronel, **Francisco Enrique Vicente Castro**, Anna Christine Amarra, Rose Ann Camille Caliso, and John Paul Vergara. eHealth TABLET: A Developing Country Perspective in Managing the Development and Deployment of a Mobile - Cloud Electronic Medical Record for Local Government Units. In *2014 IEEE 15th International Conference on Mobile Data Management (MDM)*, volume 1, pages 313–316, July 2014. DOI: [10.1109/MDM.2014.44](https://doi.org/10.1109/MDM.2014.44) (PDF).

Book Chapters

- 2013 Ma Regina Estuar, Dennis Batangan, Andrei Coronel, Anna Christine Amarra, and **Francisco Castro**. Bottom Up Approach and Devolved Design of a Health Information System: eHealth TABLET. In Kazayuki Imamura, Shiro Usui, Tomoaki Shirao, Takuji Kasamatsu, Lars Schwabe, and Ning Zhong, editors, *Brain and Health Informatics*, number 8211 in Lecture Notes in Computer Science, pages 210–217. Springer International Publishing, October 2013. DOI: [10.1007/978-3-319-02753-1_21](https://doi.org/10.1007/978-3-319-02753-1_21) (PDF).

Refereed Research Posters

- 2025 **Francisco Enrique Vicente Castro**, Mohammad Hadi Nezhad, Mark Iyer, and Ivon Arroyo. Using open-ended responses to design an assessment tool for ethical and responsible computing learning activities. In *Proceedings of the 25th Koli Calling International Conference on Computing Education Research*, Koli Calling '25, New York, NY, USA, 2025. Association for Computing Machinery. DOI: [10.1145/3769994.3770060](https://doi.org/10.1145/3769994.3770060) (PDF).
- 2024 **Francisco Enrique Vicente G. Castro** and Kayla DesPortes. Supporting the Development of AI Literacy Competencies Within Creative Computing Environments. In *Proceedings of the 2024 ACM Conference on International Computing Education Research - Volume 2*, ICER '24, page 529–530, New York, NY, USA, August 2024. Association for Computing Machinery. DOI: [10.1145/3632621.3671422](https://doi.org/10.1145/3632621.3671422) (PDF).
- 2023 **Francisco Castro**, Sangho Suh, Jane L. E, Weena Naowaprateep, and Yang Shi. Developing comic-based learning toolkits for teaching computing to elementary school learners. In *Proceedings of the 54th ACM Technical Symposium on Computer Science Education V. 2*, SIGCSE 2023, page 1325, New York, NY, USA, 2023. Association for Computing Machinery. DOI: [10.1145/3545947.3576272](https://doi.org/10.1145/3545947.3576272) (PDF).
- 2022 **Francisco Enrique Vicente Castro**, Kayla DesPortes, William Payne, Yoav Bergner, and Kathleen McDermott. AI + Dance: Co-Designing Culturally Sustaining Curricular Resources for AI and Ethics Education Through Artistic Computing. In *Proceedings of the 2022 ACM Conference on International Computing Education Research - Volume 2*, ICER '22, pages 26–27, New York, NY, USA, August 2022. Association for Computing Machinery. DOI: [10.1145/3501709.3544275](https://doi.org/10.1145/3501709.3544275) (PDF).
- 2022 **Francisco Enrique Vicente Castro**. Exploring the Use of Finite-State Machines and Game Creation to Teach Computational Thinking in Middle Schools. In *Proceedings of the 27th ACM Conference on on Innovation and Technology in Computer Science Education Vol. 2*, ITiCSE '22, page 618, New York, NY, USA, July 2022. Association for Computing Machinery. DOI: [10.1145/3502717.3532137](https://doi.org/10.1145/3502717.3532137) (PDF).
- 2021 **Francisco Enrique Vicente Castro** and Minji Kong. Integrating the Analytics of Student Interaction Data Within Scratch with a Programming Skills Taxonomy. In *Proceedings of the 17th ACM Conference on International Computing Education Research*, pages 431–432. Association for Computing Machinery, New York, NY, USA, August 2021. DOI: [10.1145/3446871.3469788](https://doi.org/10.1145/3446871.3469788) (PDF).
- 2021 Juho Leinonen, **Francisco Enrique Vicente Castro**, and Arto Hellas. Fine-Grained Versus Coarse-Grained Data for Estimating Time-on-Task in Learning Programming. In *Proceedings of the 14th International Conference on Educational Data Mining*. International Educational Data Mining Society, June 2021. (EDM 2021 – <https://educationaldatamining.org/EDM2021/EDM2021Proceedings.pdf>).

- 2019 **Francisco Enrique Vicente Castro** and Kathi Fisler. Balancing Act: A Theory on the Interactions Between High-Level Task-thinking and Low-Level Implementation-thinking of Novice Programmers. In *Proceedings of the 2019 ACM Conference on International Computing Education Research*, ICER '19, New York, NY, USA, 2019. ACM. DOI: [10.1145/3291279.3341204](https://doi.org/10.1145/3291279.3341204) (PDF).
- 2018 **Francisco Enrique Vicente G. Castro**. Towards a Theory of HtDP-based Program-Design Learning. In *Proceedings of the 2018 ACM Conference on International Computing Education Research*, ICER '18, pages 260–261, New York, NY, USA, 2018. ACM. DOI: [10.1145/3230977.3231020](https://doi.org/10.1145/3230977.3231020) (PDF).
- 2016 **Francisco Enrique Vicente G. Castro**. Pedagogy and Measurement of Program Planning Skills. In *Proceedings of the 2016 ACM Conference on International Computing Education Research*, ICER '16, pages 273–274, New York, NY, USA, 2016. ACM. DOI: [10.1145/2960310.2960344](https://doi.org/10.1145/2960310.2960344) (PDF).
- 2015 **Francisco Enrique Vicente G. Castro**. Investigating Novice Programmers' Plan Composition Strategies. In *Proceedings of the Eleventh Annual International Conference on International Computing Education Research*, ICER '15, pages 249–250, New York, NY, USA, 2015. ACM. DOI: [10.1145/2787622.2787735](https://doi.org/10.1145/2787622.2787735) (PDF).

Refereed Panels, Workshops, and Workshop Papers

- 2026 Sri Yash Tadimalla, **Francisco Enrique Vicente Castro**, Stephanie T. Jones, Corina Hernandez, F. Megumi Kivuva, Amy J. Ko, and Wendy M. Dubow. From belonging to action: Advancing lgbtq+ inclusion in computing education. In *Proceedings of the 57th ACM Technical Symposium on Computer Science Education V.2*, SIGCSE TS 2026, page 1708, New York, NY, USA, 2026. Association for Computing Machinery. DOI: [10.1145/3770761.3777098](https://doi.org/10.1145/3770761.3777098) (PDF).
- 2025 F. Megumi Kivuva, Joslenne Peña, **Francisco Enrique Vicente Castro**, Michael Miljanovic, and Amy J. Ko. Building and sustaining queer communities in computing education: Activism, creativity, and connection. In *Proceedings of the 56th ACM Technical Symposium on Computer Science Education V. 2*, SIGCSETS 2025, page 1720, New York, NY, USA, 2025. Association for Computing Machinery. DOI: [10.1145/3641555.3705093](https://doi.org/10.1145/3641555.3705093) (PDF).
- 2024 Joslenne Peña, F. Megumi Kivuva, Mara Kirdani-Ryan, **Francisco Enrique Vicente Castro**, and Amy J. Ko. Cultivating and Celebrating LGBTQ+ Community in Computing Education. In *Proceedings of the 55th ACM Technical Symposium on Computer Science Education V. 2*, SIGCSE 2024, page 1919, New York, NY, USA, 2024. Association for Computing Machinery. DOI: [10.1145/3626253.3635387](https://doi.org/10.1145/3626253.3635387) (PDF).
- 2024 Amy J. Ko, **Francisco Castro**, Aakash Gautam, Anne Drew Hu, Sara Kingsley, Michael Lachney, and Aman Yadav. Computing, Education, and Capitalism. In *Proceedings of the 55th ACM Technical Symposium on Computer Science Education V. 2*, SIGCSE 2024, page 1914, New York, NY, USA, 2024. Association for Computing Machinery. DOI: [10.1145/3626253.3635362](https://doi.org/10.1145/3626253.3635362) (PDF).
- 2023 Benjamin Xie, Greg L. Nelson, **Francisco Enrique Vicente Castro**, Nicholas Lytle, and Briana Bettin. Centering environmental justice in computing education. In *Proceedings of the 54th ACM Technical Symposium on Computer Science Education V. 2*, SIGCSE 2023, page 1244, New York, NY, USA, 2023. Association for Computing Machinery. DOI: [10.1145/3545947.3573343](https://doi.org/10.1145/3545947.3573343) (PDF).
- 2023 **Francisco Enrique Vicente Castro**, Joseph Wilson, Jessica Vandenberg, Juho Leinonen, and Miranda C. Parker. Computing education postdocs and beyond: Building a postdoc space for community and collaboration. In *Proceedings of the 54th ACM Technical Symposium on Computer Science Education V. 2*, SIGCSE 2023, page 1247, New York, NY, USA, 2023. Association for Computing Machinery. DOI: [10.1145/3545947.3573340](https://doi.org/10.1145/3545947.3573340) (PDF).

- 2022 Benjamin Walsh, Safinah Ali, **Francisco Castro**, Kayla DesPortes, Daniella DiPaola, Irene Lee, William Payne, Scott Sieke, and Helen Zhang. Making Art with and about Artificial Intelligence: Three Approaches to Teaching AI and AI Ethics to Middle and High School Students. In *Proceedings of the 53rd ACM Technical Symposium on Computer Science Education V. 2*, SIGCSE 2022, page 1203, New York, NY, USA, March 2022. Association for Computing Machinery. DOI: [10.1145/3478432.3499157](https://doi.org/10.1145/3478432.3499157) (PDF).
- 2022 **Francisco Enrique Vicente Castro**, Kathryn Cunningham, Miranda Parker, and Nicholas Lytle. What's Up, Doc? Building a Community of Computing Education Postdocs. In *Proceedings of the 53rd ACM Technical Symposium on Computer Science Education V. 2*, SIGCSE 2022, page 1186, New York, NY, USA, March 2022. Association for Computing Machinery. DOI: [10.1145/3478432.3499173](https://doi.org/10.1145/3478432.3499173) (PDF).
- 2021 **Francisco Castro**, Earl Huff, Gayithri Jayathirtha, Yerika Jimenez, Minji Kong, Natalie Araujo Melo, Amber Solomon, and Jennifer Tsan. What Are Your Papayas? Assimilating to Belong in Computing. In *CMD-IT/ACM Richard Tapia Celebration of Diversity in Computing*, Tapia Conference 2021, September 2021. [Tapia Conference Workshop: Broadening Participation in Computing](https://www.tapiaconf.org/).
- 2021 **Francisco Castro**, Earl Huff, Gayithri Jayathirtha, Yerika Jimenez, Minji Kong, Natalie Araujo Melo, Amber Solomon, and Jennifer Tsan. Examining and Redesigning Computing Education Research to Center Equity. In *International Computing Education Research Conference, ICER '21*. Association for Computing Machinery, August 2021. [ICER 2021 Co-located workshop – https://icer2021.acm.org/info/co-located-workshops](https://icer2021.acm.org/info/co-located-workshops).
- 2021 Earl Huff, **Francisco Castro**, Gayithri Jayathirtha, Yerika Jimenez, Minji Kong, Natalie Araujo Melo, Amber Solomon, and Jennifer Tsan. Going Through a Process of Whitening: Student Experiences Within Computer Science Education. In *Proceedings of the 52nd ACM Technical Symposium on Computer Science Education*, SIGCSE '21, page 1348, New York, NY, USA, March 2021. Association for Computing Machinery. DOI: [10.1145/3408877.3432497](https://doi.org/10.1145/3408877.3432497) (PDF).

Technical Reports

- 2026 Mohammad Hadi Nezhad, **Francisco Enrique Vicente Castro**, and Ivon Arroyo. Understanding Users' Privacy Reasoning and Behaviors During Chatbot Use to Support Meaningful Agency in Privacy. Technical report, 2026. DOI: [10.48550/arXiv.2601.18125](https://doi.org/10.48550/arXiv.2601.18125) (PDF).
- 2026 Mohammad Hadi Nezhad, **Francisco Enrique Vicente Castro**, and Ivon Arroyo. Investigating In-Context Privacy Learning by Integrating User-Facing Privacy Tools into Conversational Agents. Technical report, 2026. DOI: [10.48550/arXiv.2603.19416](https://doi.org/10.48550/arXiv.2603.19416) (PDF).
- 2025 Kenya S. Andrews, Deborah Dormah Kanubala, Kehinde Aruleba, **Francisco Enrique Vicente Castro**, and Renata A Revelo. A Justice Lens on Fairness and Ethics Courses in Computing Education: LLM-Assisted Multi-Perspective and Thematic Evaluation, 2025. DOI: [10.48550/arXiv.2510.18931](https://doi.org/10.48550/arXiv.2510.18931) (PDF).
- 2017 **Francisco Enrique Vicente Castro**, Shriram Krishnamurthi, and Kathi Fisler. The Impact of a Single Lecture on Program Plans in First-year CS. Technical Report WPI-CS-TR-17-02, Worcester Polytechnic Institute, Computer Science Department 100 Institute Road, Worcester, Massachusetts 01609-2280, November 2017. (PDF – <https://ftp.cs.wpi.edu/pub/techreports/pdf/17-02.pdf>).

In Press/In Revision/Under Review

- In revision **Francisco Enrique Vicente Castro**, Mohammad Hadi Nezhad, Mark Iyer, Danielle Allesio, and Ivon Arroyo. Developing a Framework of CS Students' Thinking on Responsible Computing. In revision. [In revision](#).

In revision Tsvetomila Mihaylova, Evanfiya Logacheva, Arto Hellas, Jing Fan, **Francisco Enrique Vicente Castro**, Bitá Akram, Narges Norouzi, Peter Brusilovsky, and Juho Leinonen. The Effects of Structured LLM-Generated Feedback on Programming Assignment Performance. In revision. [In revision](#).

Tools and Platforms - Research

Creative Computing Cookbook, <https://creativecomputingcookbook.github.io/cwp-11ty/>, Funded through US NSF ITEST 2241809.

Platform for Ethical and Responsible Computing Education, <https://pearce-sim.org/>, Funded through US NSF IUSE 2337133 and 2337132.

danceON - Dance Object Notation, <https://dancinganddabbing.github.io/danceON/>, Funded through US NSF STEM+C 1933961.

Data Remixer, <https://data-remixer-dev-server.onrender.com/>, Funded through US NSF Cyberlearn 2302658.

WearableLearning.org, <http://wearablelearning.org/>, Funded through US NSF STEM+C 2026722 and 2041785.

Grants, Fellowships, and Funding Awarded

Current/Ongoing Grants

2024 – 2026 **US National Science Foundation Grant 2337133**, (*NSF IUSE 2337133*) *Development and Impact Assessment of an Interactive Online System for Computing Ethics Education*, \$98,753.00 USD, https://www.nsf.gov/awardsearch/showAward?AWD_ID=2337133.

○ PI: **Francisco Castro**

Fellowships Awarded

2025 **Aalto Science Institute (ASCI) - Visiting Researcher programme**, *Investigating the Use of Generative AI to Support Data Literacy*, Aalto University, <https://www.aalto.fi/en/aalto-science-institute-asci/visiting-scientist-programmes>.

2022 – 2024 **Computing Innovation Fellowship - Computing Research Association (CRA), National Science Foundation (NSF), Computing Community Consortium (CCC)**, *Investigating the Design of Embodied AI Ethics Education in Dance Computing Spaces*, \$225,000 USD, cifellows2021.org.

2023 – 2024 **Quantitative Ethnography (QE) Fellows**, *QE Fellows Institute*, qefellows.org/qe-fellows/.

2021 – 2023 **Cultural Competence in Computing**, *3C Fellows Program*, identity.cs.duke.edu/fellows.

Completed Grants

2023 – 2024 **New York University Postdoctoral Research and Professional Development Support Grants**, *AI + Art: Culturally Sustaining AI and Ethics Education Through Artistic and Creative Practice*, \$2,390 USD.

○ PIs: **Francisco Enrique Vicente Castro**, Kayla DesPortes

2023 – 2024 **International Society of the Learning Sciences Emerging Scholars Program (Wallace Foundation)**, *ML + Dance: Co-Designing Culturally Sustaining Curricular Resources*, \$10,000.00 USD, *ISLS Emerging Scholars*, <https://www.isls.org/about/isls-emerging-scholars/#fcastro>.

○ PI: **Francisco Castro**

- 2022 **Helsinki Institute for Information Technology (HIIT)/Finnish Center for Artificial Intelligence (FCAI)**, *Strengthening Collaboration Between Aalto University and New York University in Computing Education Research*, €2,443.80 EUR, hiit.fi.
 o Pls: Arto Hellas, **Francisco Enrique Vicente Castro**, Juho Leinonen
- 2021 – 2023 **ACM SIGCSE Special Projects Grant**, *Telling Our Narratives: Expanding Equity Within Computing Education*, \$4,098 USD, sigcse.org.
 o PI: **Francisco Enrique Vicente Castro**

Travel Grants

- 2023 **International Society of the Learning Sciences Travel Fund**, \$430 USD.
- 2018 **ACM International Computing Education Research Travel Grant**, *Towards a Theory of HtDP-based Program-Design Learning*, \$500 USD.
- 2016 **ACM International Computing Education Research Travel Grant**, *Pedagogy and Measurement of Program Planning Skills*, \$500 USD.
- 2015 **ACM International Computing Education Research Travel Grant**, *Investigating Novice Programmers' Plan Composition Strategies*, \$500 USD.

Service and Professional Activities

Invited Talks and Lectures

- 2023 **Empowering Learners and Educators Through AI and Creative Computing**, *Francisco Castro, PhD*, Keynote Talk: All EmpowerEd 2023 School Principals and Counselors Conference, (November 11, 2023) Jakarta, Indonesia – [OSF: Keynote Talk, https://osf.io/rqf5g](https://osf.io/rqf5g).
- 2023 **AI and Education Equity in Higher Education**, *Francisco Castro, Erin Walker, Kenya Andrews, Jaemarie Solyst, Benjamin Xie, Lili Yan*, YouTube Panel Talk: Empowering Learners for the Age of AI (ELAI) 2023 Conference, (October 25, 2023) – <https://www.youtube.com/watch?v=GghDbkumBSg>.
- 2023 **Empowering Learners Through AI, Dance, and Creative Computing**, *Francisco Castro, PhD*, University of Pennsylvania, Invited Talk: Artificial Intelligence for Children and Youth: Learning, Creating, and Understanding (June 22, 2023).
- 2022 **Empowering Learners and Educators Through Creative Computing**, *Francisco Castro, PhD*, Aalto University Invited Talk: Department of Computer Science, (November 24, 2022) – <https://www.aalto.fi/en/events/guest-talk-francisco-castro-nyu-empowering-learners-and-educators-through-creative-computing>.
- 2022 **Recipes for ResistanCSe - Identity & Computing Lecture Series**, *Dr. Francisco Castro, Dr. Earl Huff Jr., Dr. Gayithri Jayathirtha, Yerika Jimenez, Minji Kong, Natalie Araujo Melo, Dr. Amber Solomon, Dr. Jennifer Tsan, Dr. Briana Bettin, Victoria Chavez*, Alliance for Identity Inclusive Computing Education (AIICE), URL: <https://identityincs.org/event/identity-computing-lecture-series-recipes-resistancse/> (November 15, 2022).
- 2022 **Making CS Education More Equitable and Accessible for Our Most Marginalized Students**, *Dr. Francisco Castro, Dr. Jennifer Tsan, Victoria Chavez, Natalie Araujo Melo*, ACM Special Interest Group in Computers and Society Special Edition Vision in Progress Talk Series, URL: <https://www.sigcas.org/2022/05/31/june-8-2022-making-cs-education-more-equitable-and-accessible-for-our-most-marginalized-students/>.
- 2017 **A Data-Centric Introduction to Programming – Designing Programs**, *Francisco Castro*, Brown University Department of Computer Science, Invited Lecture: CSCI0050.

Initiatives for Equity and Diversity in Computing

- 2020 – present **Co-Founder, The Papaya Project**, the-papaya-project.github.io.
The Papaya Project is an initiative towards identifying and critically addressing inequities, discrimination, and bias in computing and computing education research, with the goal of transforming computing education and the broader computing discipline into a more inclusive, diverse, and equitable field.
◦ **Founders:** **Francisco Castro, PhD**, Earl Huff Jr., PhD, Gayithri Jayathirtha, PhD, Yerika Jimenez, PhD, Minji Kong, Natalie Araujo Melo, PhD, Amber Solomon, PhD, Jennifer Tsan, PhD
- 2022 – present **Host, Recipes for Resistance**, recipes4resistance.github.io/.
The Recipes for Resistance Podcast is a multi-modal podcast series that provides a platform for minoritized voices to discuss how to foster equity in the CS and CS Education (CSE) fields. The podcast seeks to elevate voices and ideas beyond those traditionally centered, including CS teachers, parents, students, and community members whose lived experiences include doing work by and for their communities. Our podcast episodes are crafted around a kitchen table talk, where we informally discuss the topic of choice and share stories, all while centering justice, joy, and healing.

Research Conference Organizing Committee/Program Committee

- AIED **International Conference on Artificial Intelligence in Education**, *International Artificial Intelligence in Education Society (IAIED)*.
◦ **AIED Program Committee** — **2025** (University of Palermo, Italy) | **2024** (Recife, Brazil) | **2023** (Hitotsubashi University, Tokyo, Japan) | **2022** (University of Durham, United Kingdom) |
- EAAI **Symposium on Educational Advances in Artificial Intelligence**, *Association for the Advancement of Artificial Intelligence*.
◦ **EAAI Program Committee** — **2026** (Singapore) | **2024** (Vancouver, Canada) | **2022** (Washington, DC, USA) |
- FabLearn **FabLearn**, *Association for Computing Machinery*.
◦ **FabLearn/Constructionism Track Chair** — **2023 Posters Track Chair** (New York, New York, USA)
- SIGCSE **ACM Technical Symposium on Computer Science Education**, *Association for Computing Machinery*.
◦ **SIGCSE Ethics Facilitator** — **2023 Ethics Facilitator**
◦ **SIGCSE Program Committee** — **2024** (Portland, Oregon, United States) | **2023** (Toronto, Canada) | **2022** (Providence, Rhode Island, United States) | **2021** (Online) | **2020** (Online/Portland, Oregon, United States) | **2019** (Minneapolis, Minnesota, United States) | **2018** (Baltimore, Maryland, United States) | **2017** (Seattle, Washington, United States)
- ITiCSE **Conference on Innovation and Technology in Computer Science Education**, *Association for Computing Machinery*.
◦ **ITiCSE Program Committee** — **2023** (Turku, Finland) | **2022** (Dublin, Ireland) | **2021** (Paderborn, Germany) | **2020** (Trondheim, Norway) | **2019** (Aberdeen, UK) | **2018** (Larnaca, Cyprus) | **2017** (Bologna, Italy)
- ICER **International Computing Education Research Conference**, *Association for Computing Machinery*.
◦ **ICER Ethics Facilitator** — **2022 Ethics Facilitator**
◦ **ICER Program Committee** — **2025** (Charlottesville, Virginia, United States) | **2024** (Melbourne, Australia) | **2023** (Chicago, Illinois, United States) | **2022** (Lugano, Switzerland)
- IDC **ACM Interaction Design and Children Conference**, *Association for Computing Machinery*.
◦ IDC 2021, Athens, Greece
- DSE-K12 **Data Science Education K-12: Research to Practice Conference**, *Data Science 4 Everyone and The Concord Consortium*.
◦ **DSE-K12 Program Committee and Inclusivity and Accessibility Working Group** — **2025** (San Antonio, Texas, United States)

Koli Calling 2020 – present **Koli Calling International Conference on Computing Education Research**, *Association for Computing Machinery*, Koli, Finland.

RESPECT **Conference on Research in Equity and Sustained Participation in Engineering, Computing, and Technology**, *Association for Computing Machinery*, Institute of Electrical and Electronics Engineers (Previous).

- RESPECT Program Committee — **2025 Research Track Chair** (New Jersey Institute of Technology, Newark, New Jersey)

- RESPECT Program Committee — **2022** (Philadelphia, Pennsylvania, United States) | **2021** (Virtual)

CompEd **ACM Global Computing Education Conference**, *Association for Computing Machinery*.

- CompEd Program Committee — **2019** (Chengdu, China)

PCSC **Philippine Computing Science Congress**, *Computing Society of the Philippines*.

- PCSC Program Committee — **2017** (Cebu, Philippines) | **2016** (Palawan, Philippines) |

Research Journals - Reviewer

IJAIED **International Journal of Artificial Intelligence in Education**, *Springer Nature*, <https://link.springer.com/journal/40593>.

ACM-JRC **ACM Journal on Responsible Computing**, *Association for Computing Machinery*, <https://dl.acm.org/journal/jrc>.

IJCCI **International Journal of Child-Computer Interaction**, *Elsevier*, <https://www.sciencedirect.com/journal/international-journal-of-child-computer-interaction>.

CSE **Computer Science Education**, *Taylor & Francis*, <https://www.tandfonline.com/journals/ncse20>.

ACM-TOCE **ACM Transactions on Computing Education**, *Association for Computing Machinery*, <https://dl.acm.org/journal/toce>.

EJEE **European Journal of Engineering Education**, *European Society for Engineering Education*, *Taylor & Francis*, <https://www.tandfonline.com/journals/ceee20>.

EngageCSEdu **EngageCSEdu**, *Association for Computing Machinery*, *National Center for Women & Information Technology (NCWIT)*, <https://www.engage-csedu.org/>.

ACM Inroads **ACM Inroads**, *Association for Computing Machinery (ACM)*, <https://inroads.acm.org/>.

Advisory Board

2023 – 2024 **Center for Integrative Research in Computing and Learning Sciences and Journal of Computing in Higher Education**, URL: <https://link.springer.com/article/10.1007/s12528-025-09481-1>, Special Issue Call for Papers: Equity of Artificial Intelligence in Higher Education, URL: <https://circls.org/ai-circls-and-jche-collaboration>.
AI CIRCLS and JCHE Collaboration

US National Science Foundation Review Panels

2024 **Division of Research on Learning in Formal and Informal Settings (EDU/DRL)**, *Racial Equity in STEM Education*.

2021 **Division of Research on Learning in Formal and Informal Settings (EDU/DRL)**, *Innovative Technology Experiences for Students and Teachers*.

Research Panels and Discussion Leader

- 2022 **Making Art with and about Artificial Intelligence: Three Approaches to Teaching AI and AI Ethics to Middle and High School Students**, *SIGCSE 2022 ACM Technical Symposium on Computer Science Education*.
- 2022 **What's Up, Doc? Building a Community of Computing Education Postdocs**, *SIGCSE 2022 ACM Technical Symposium on Computer Science Education*.
- 2021 **Going Through a Process of Whitening: Student Experiences Within Computer Science Education**, *SIGCSE 2021 ACM Technical Symposium on Computer Science Education*.
- 2016 **Birds of a Feather Session: CS Ed PhD Students Unite!**, *SIGCSE 2016 ACM Technical Symposium on Computer Science Education*.

University Committees

- 2016 – 2018 **Vice President**, *Worcester Polytechnic Institute Undergraduate Research Journal*.
- 2014 – 2016 **Graduate Council Member**, *Worcester Polytechnic Institute Computer Science Department*.
- 2014 – 2015 **Senate Representative - Computer Science**, *Worcester Polytechnic Institute Graduate Student Government Senate*.

Community Service and Outreach

- 2015 – 2017 **Staff**, *WPI TouchTomorrow: A Festival of Science, Technology, and Robots*, Worcester Polytechnic Institute.

Honors, Awards, and Recognitions

- 2025 **Steinhardt Distinguished Professional Researcher**, *New York University*, New York, NY USA.
- 2022 **Computing Education Research Best Paper Award**, *Time-on-Task for Predicting Performance*, 53rd ACM Technical Symposium on Computer Science Education, Providence, RI, USA.
- 2017 **Best Paper Award**, *Designing a Multi-Faceted SOLO Taxonomy to Track Program Design Skills Through an Entire Course*, 17th Koli Calling International Conference on Computing Education Research, Koli, Finland.
- 2014 **Teaching Assistant of the Year**, *Department of Computer Science*, Worcester Polytechnic Institute, Worcester, MA, USA.
- 2012 **Best Paper Award**, *Microsimulations of Arching, Clogging, and Bursty Exit Phenomena in Crowd Dynamics*, 10th National Conference on Information Technology Education, Philippines.

Press and Media Coverage

- 2025 **Unpacking Colonial Histories and Monolithic Perspectives of Asian Identities Within STEM and Computing Education**, *Recipes for Resistance Podcast*.
- 2024 **CICS Research Team Receives NSF Grant for Online Interactive Platform for Ethical and Responsible Computing Education**, *University of Massachusetts Amherst - Manning College of Information and Computer Sciences*, ([URL: https://www.umass.edu/news/article/cics-research-team-receives-nsf-grant-online-interactive-platform-ethical-and](https://www.umass.edu/news/article/cics-research-team-receives-nsf-grant-online-interactive-platform-ethical-and)).
University of Massachusetts Amherst News
- 2022 **ReCSistance: Framing the conversation on resistance**, *Recipes for Resistance Podcast*, ([URL: https://recipes4resistance.github.io/2022/09/15/e1p1.html](https://recipes4resistance.github.io/2022/09/15/e1p1.html)).
Recipes for Resistance Podcast Episode

- 2022 **Brain, Body, World**, University of Massachusetts Amherst College of Information & Computer Sciences News, (URL: <https://www.cics.umass.edu/news/brain-body-world>).
UMass Amherst College of Information and Computer Sciences (CICS) News Article
- 2022 **Brain, Body, World**, University of Massachusetts Amherst College of Education News, (URL: <https://www.umass.edu/education/stories/brain-body-world>).
Brain, Body, World — UMass Amherst College of Education News Article
- 2022 **Francisco Castro: Empowering People Through Creative Computing**, Center for Integrative Research in Computing and Learning Sciences (CIRCLS) Featured Emerging Scholars, (URL: <https://circls.org/emerging-scholar/francisco-castro-empowering-people-through-creative-computing>).
- 2021 **Embodying Computational Thinking and Math with Mobile and Wearable Learning Technologies**, 2021 STEM for All Video Showcase, (URL: <https://stemforall2021.videohall.com/presentations/2164.html>).
- 2020 **Developing Computational Thinking Through Game Design & Play**, 2020 STEM for All Video Showcase, (URL: <https://stemforall2020.videohall.com/presentations/1711.html>).
- 2019 **WPI Researchers Awarded Grant to Help K-12 Students Design Math Games**, Worcester Polytechnic Institute Research News, (URL: <https://www.wpi.edu/news/wpi-researchers-awarded-grant-help-k-12-students-design-math-games>).
Worcester Polytechnic Institute Research News at wpi.edu
- 2019 **WPI researchers get \$746K to develop math game site**, Worcester Business Journal, (URL: <https://www.wbjournal.com/article/wpi-researchers-get-746k-to-develop-math-game-site>).
Worcester Business Journal - wbjournal.com

Professional Societies and Organizations

Association for Computing Machinery (ACM).

Association for Computing Machinery (ACM), *Special Interest Group on Computer Science Education*.

Association for Computing Machinery (ACM), *Special Interest Group on Computer-Human Interaction*.

Out in Science, Technology, Engineering, and Mathematics (oSTEM).

International Society of the Learning Sciences (ISLS).

American Educational Research Association (AERA).

Society for Learning Analytics Research (SoLAR).

Computer Science Teachers Association (CSTA).

National Science Policy Network (NSPN).

Student Mentorship and Advising

Doctoral Committee

2023 **Marco Sbaraglia**, *Computer Science and Engineering*, Università di Bologna.

2021 – 2025 **Sai Satish Gattupalli**, *Learning Technologies*, University of Massachusetts Amherst.

Early Research Scholars Program (ERSP) UMass Amherst

2022 – 2023 **Julia Epshtein**, *Computer Science*, University of Massachusetts Amherst.

2022 – 2023 **Anh Pham**, *Computer Science*, University of Massachusetts Amherst.

2022 – 2023 **Minh Vu**, *Computer Science*, University of Massachusetts Amherst.

2021 – 2022 **Varun Gandhi**, *Computer Science*, University of Massachusetts Amherst.

2021 – 2022 **Vivien Jamba**, *Computer Science*, University of Massachusetts Amherst.

- 2021 – 2022 **Foram Patel**, *Computer Science*, University of Massachusetts Amherst.
2021 – 2022 **Vivienne Tam**, *Computer Science*, University of Massachusetts Amherst.

Graduate Research

- 2025 – present **Vinayak Singh Bhadoriya**, New York University.
2025 **Austin Huang**, New York University.
2024 **Chengsheng Zhao**, New York University.
2024 – present **Sabria Islam**, New York University.
2023 – present **Cassandra Scheirer**, New York University.
2023 – 2024 **Shuang Cai**, New York University.
2023 – 2024 **Vera Zhong**, New York University.
2023 – present **Mohammad Hadi Nezhad**, *Computer Science*, University of Massachusetts Amherst.
2021 – 2023 **Allison Poh**, *Computer Science*, University of Massachusetts Amherst.
2018 **Jun Rangie Obispo**, *Computer Science*, Ateneo de Manila University.

Undergraduate Research

- 2025 – present **Mark Iyer**, *Computer Science*, University of Massachusetts Amherst.
2025 – present **Billie Chen**, New York University.
2016 – 2017 **William Temple**, *Computer Science*, Worcester Polytechnic Institute.
2013 – 2014 **Michael Ignacio**, *Computer Science*, University of the Philippines Los Baños.
2014 **Luke Prudente**, *Computer Science*, Ateneo de Manila University.

K-12 and Afterschool Programs

- 2022 – present **CWP 2.0 High School Students**, *Community Word Project and New York University*, New York City, communitywordproject.org.